

Use Case ID & Name	UC-03: Place Order	System Scope	Multi-Vendor Artisan Marketplace
Primary Actor	Shopper	Secondary Actors	Payment Gateway, Artisan Vendor
Trigger	Shopper clicks "Proceed to Checkout" from cart	Included / Extended UCs	< <include>> UC-04 (Split Pay), < <extend>> UC-05 (Discount)

PRECONDITIONS

- Shopper is authenticated with valid shipping address saved.
- Shopping cart contains items from ≥ 1 active artisan vendor storefronts.
- Item inventory quantities are verified available in catalog.
- Vendors have active, verified payout disbursement accounts.

POSTCONDITIONS

- Master order created and decomposed into vendor sub-orders.
- Consolidated cart payment authorized by Payment Gateway.
- **5% platform fee deducted; net payouts split to vendor escrows (FR-001).**
- Product inventory counts decremented; confirmation emails sent.

MAIN SUCCESS SCENARIO

1. Shopper navigates to the checkout view from the multi-vendor shopping cart.
2. System retrieves cart items across all artisan vendors, locks inventory quantities, and displays an itemized order summary.
3. Shopper selects or confirms the delivery shipping address and recipient contact details.
4. System computes shipping costs per artisan vendor, applicable taxes, and presents the grand total payable amount.
5. Shopper selects payment method (Credit/Debit Card, UPI, Net Banking) and enters payment credentials.
6. System invokes <<include>> **UC-04: Process Split Payment**, routing encrypted transaction payload (TLS 1.3) to the **Payment Gateway**.
7. Payment Gateway authorizes the full consolidated cart amount and returns a success authorization token.
8. System automatically **deducts the 5% platform commission** and allocates remaining item earnings to respective artisan vendor payout accounts (adhering to **FR-001**).
9. System decrements catalog inventory quantities, decomposes master order into isolated vendor sub-orders, and schedules disbursement settlements.
10. System displays an order confirmation screen with order ID and transmits confirmation receipts to the shopper and alert notifications to respective artisan vendors. Use case ends successfully.

ALTERNATE FLOWS

6a. Alternate Flow: Payment Authorization Declined (at Step 6 / 7)

- 6a1.** Payment Gateway returns an authorization failure code (e.g., insufficient funds, bank timeout, expired card).
- 6a2.** System logs the transaction attempt, flags order as 'Payment Pending', and presents a clear error message to the shopper without releasing cart items.
- 6a3.** System prompts the shopper to select an alternate payment method or re-enter valid credentials.
- 6a4.** If shopper successfully resubmits payment within 3 attempts, flow resumes at **Step 7**.
- 6a5.** If all payment attempts fail or shopper abandons session, locked inventory is released back to artisan catalogs after 15 minutes and the checkout process terminates.

4a. Alternate / Extension Flow: Apply Promotional Discount Code (<<extend>> UC-05, at Step 4)

- 4a1.** Shopper enters a promotional coupon code in the checkout discount field and clicks 'Apply'.
- 4a2.** System validates coupon eligibility against vendor items and cart criteria. If valid, system recalculates the discounted total and updates vendor payout bases; flow resumes at **Step 5**.
- 4a3.** If coupon is invalid or expired, system displays an inline warning and retains the original cart total without interrupting checkout.